

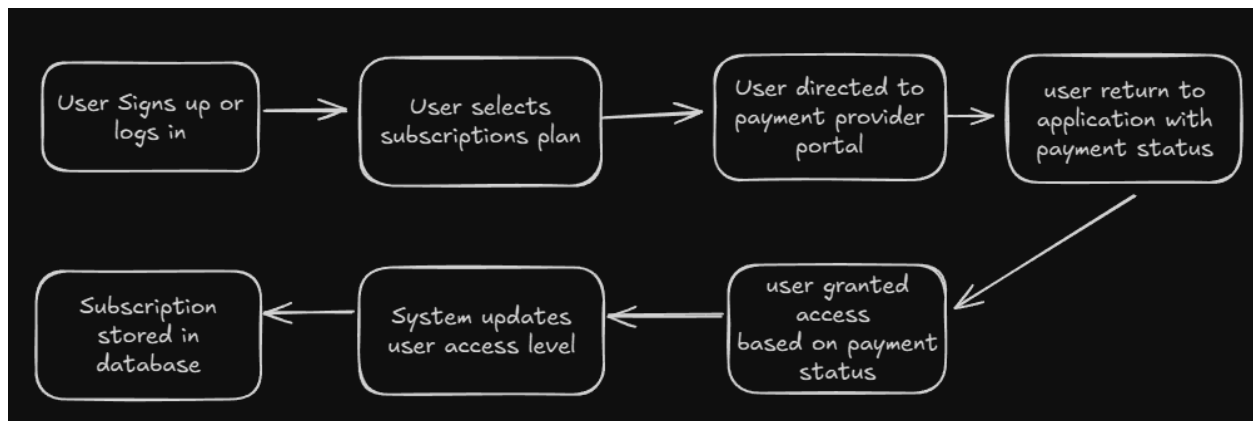
Technical Design Document: Payment Processing System for Monthly Subscriptions

Introduction

This document outlines the technical design for implementing a payment processing system to support per-user monthly subscriptions. The system will allow users to subscribe to service offerings, process payments securely, and manage access control based on subscription status.

High-Level Payment Flow

Subscription Flow



2.2 Lifecycle Events

Subscription Creation

- User selects subscription plan
- System creates pending subscription record
- User redirected to payment provider

Payment Success

- Payment provider sends webhook notification
- System validates payment
- System activates subscription
- User access level updated

Payment Failure

- Payment provider sends failure notification
- System records failed payment attempt
- User notified of payment failure
- Retry logic implemented for failed payments

Subscription Renewal

- System initiates renewal charge before expiration
- On success, subscription period extended
- On failure, retry logic activated

Subscription Cancellation

- User requests cancellation
- System marks subscription for non-renewal
- Access maintained until end of paid period

Subscription Expiration

- System detects expired subscription
- User access downgraded
- User notified of expiration

Payment Provider Selection

Recommended Provider: Stripe

Stripe is recommended for the following reasons:

- Robust API: Comprehensive, well-documented REST API
- Subscription Management: Built-in support for recurring billing
- Security Compliance: PCI DSS Level 1 certification
- Webhook Support: Real-time event notifications
- Developer-Friendly: Extensive documentation and SDKs
- Global Support: Available in 40+ countries with multi-currency support

Integration Points

Stripe Elements: For secure payment form integration

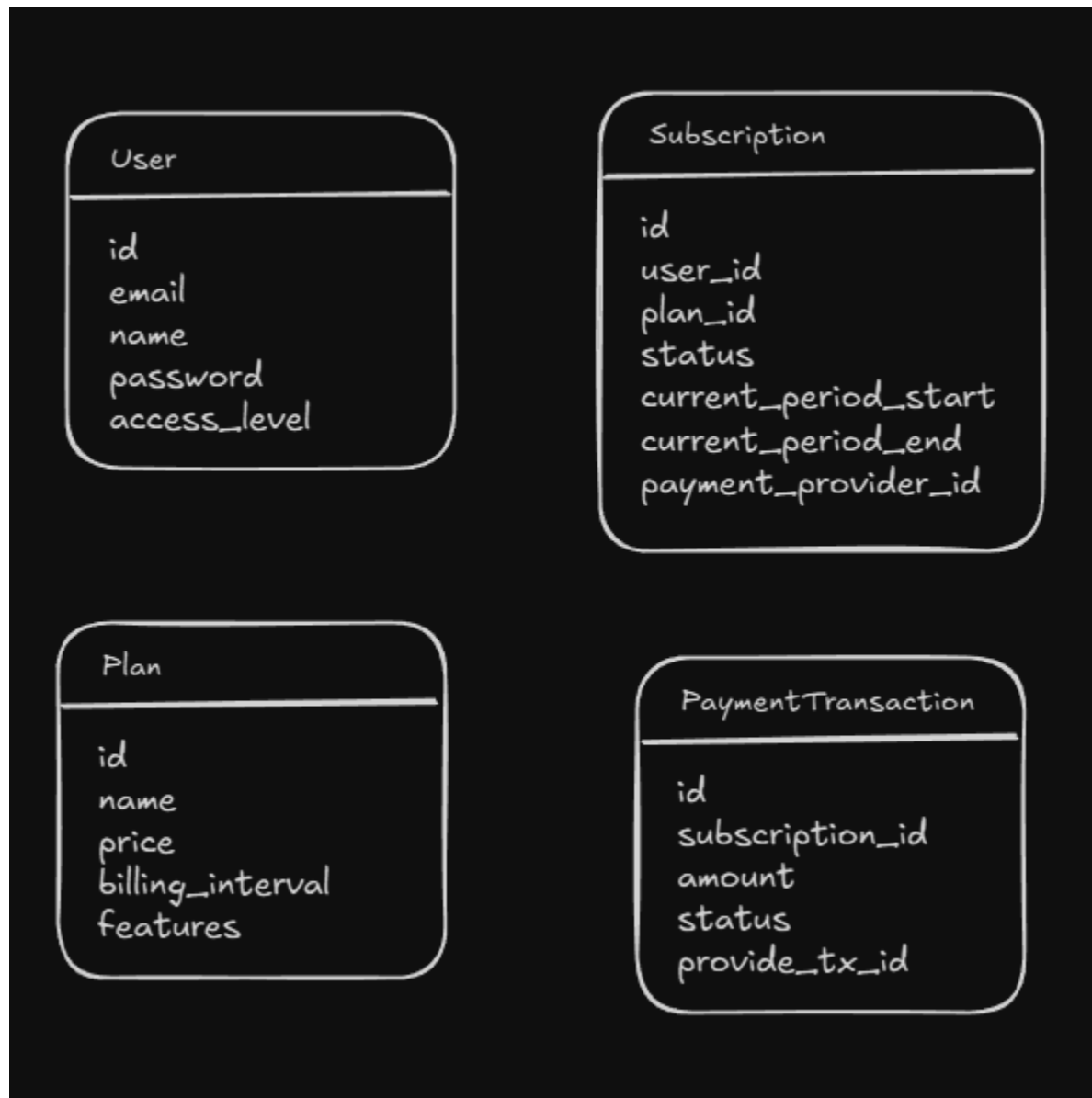
2. Stripe Checkout: For hosted payment pages
3. Stripe Customer API: For customer data management
4. Stripe Subscription API: For subscription lifecycle management
5. Stripe Webhook Events: For asynchronous event handling

Alternative Providers

- PayPal: Good for international payments, but more complex API
- Braintree: Solid option with PayPal integration
- Adyen: Strong for global markets, but higher complexity

Subscription Management Strategy

Database Structure



Subscription States

1. Incomplete: Subscription created but payment not confirmed
2. Active: Payment confirmed, subscription in good standing
3. Past Due: Payment attempted and failed, in grace period
4. Canceled: User has canceled, but period not yet ended
5. Expired: Subscription period ended without renewal

Subscription Service

The Subscription Service will handle:

- Creating new subscriptions
- Processing webhook events from payment provider
- Managing subscription state transitions
- Triggering notifications based on state changes
- Syncing subscription status with user access level

Conclusion

This design outlines a robust payment processing system for monthly subscriptions. By leveraging Stripe as the payment provider and implementing a comprehensive subscription management strategy, the system will securely handle payment processing while providing appropriate access control based on subscription status.

The solution is scalable, secure, and follows best practices for payment processing systems. With the outlined architecture, the system can be implemented incrementally, starting with core subscription functionality and expanding to more advanced features over time.